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Application-aware links

Abstract

The present technology is directed to a system and method for application aware management and recovery of link failures resulting from excessive errors observed on the link. One aspect of the proposed technology is based on identification of link errors associated with application-specific data patterns traversing link. Other aspects involve corrective actions based on relocation or modification of specific application traffic to thereby alleviate the observed excessive link errors and prevent a link failure or shut down. Relocation may involve moving the source application to a different virtual machine/container/physical device or rerouting application traffic by updating relevant routing protocols. Modification may involve harmlessly changing payload data pattern to remove data-pattern dependent signal attenuation. Information corresponding to identified faulty payload data patterns and associated frame data quality parameters maybe stored and utilized to provide analytics evaluation of network wide physical resource issues that maybe affecting application traffic.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/011,302, filed on Jun. 18, 2018, entitled APPLICATION-AWARE LINKS, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present technology pertains to improving network connectivity performance and link utilization for computer-based applications. More specifically it is directed to application level management and recovery of link failures and link analytics.

BACKGROUND

(2) All computer-implemented applications communicate through computers, networking and storage devices. The backbone interconnecting all such devices, over which application data traffic is transmitted, is comprised of plurality of physical and/or logical links. Error conditions occurring on these links are monitored and when a pre-determined error threshold on a particular link is exceeded, the link is shut down. As a consequence all data traffic traversing the link is also shut down.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the manner in which the above-recited and other advantages and features of the disclosure can be obtained, a more particular description of the principles briefly described above will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. Understanding that these drawings depict only exemplary embodiments of the disclosure and are not therefore to be considered to be limiting of its scope, the principles herein

are described and explained with additional specificity and detail through the use of the accompanying drawings in which:

- (2) FIG. 1 illustrates an exemplary transmission data frame with source application tagging, in accordance to one embodiment of the present technology.
- (3) FIG. 2 illustrates application-level delineation and error monitoring for aggregate traffic traversing an application-aware link, in accordance to one embodiment of the present technology.
- (4) FIG. 3 illustrates a basic operational flow for performing link failure identification and recovery at the application level, in accordance to one embodiment of the present technology.
- (5) FIG. 4 illustrates a relevance of application level monitoring and error management of transmission links for improving connectivity performance and link utilization, in accordance to one embodiment of the present technology.
- (6) FIG. 5 illustrates exemplary approaches directed to utilizing application tags for identifying applications-specific traffic flows, in accordance to some embodiment of the present technology.
- (7) FIG. 6 illustrates exemplary ways of modifying data frames with application tags to facilitate application-specific traffic performance monitoring, in accordance to some embodiments of the present technology.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(8) Additional features and advantages of the disclosure will be set forth in the description which follows, and in part will be obvious from the description, or can be learned by practice of the herein disclosed principles. The features and advantages of the disclosure can be realized and obtained by means of the instruments and combinations particularly pointed out in the appended claims. These and other features of the disclosure will become more fully apparent from the following description and appended claims, or can be learned by the practice of the principles set forth herein.

Overview

(9) Systems, methods, and devices are disclosed and directed to application level monitoring and analysis of error conditions on network links to identify link failures that may be attributable to specific payload data patterns associated with particular source application. Subsequent recovery actions, as disclosed by embodiments of the technology, may then be directed to modifying or relocating the specific traffic streams over to one or more different paths which would leave the network link intact for several other applications that may be transmitting over it. Aspects of the technology are further directed to storing error causing payload data patterns along with frame data quality parameters and utilizing the stored information to perform link analytics and facilitate discovery of physical resource problems impacting application traffic. It should be noted that, for the purposes of the current technology, a link generically refers to any means of providing data transport between network/computing devices. For example a link may represent an Ethernet, Optical Transport Network (OTN), Synchronous Optical Networking (SONET), Synchronous Digital Hierarchy (SDH) or any other transport protocol as would be known to a person of ordinary skill in the art.

Example Embodiments

(10) Disclosed are systems, methods, and non-transitory computer-readable storage media for application aware links capable of identifying various traffic streams traversing a link at the application level and taking subsequent corrective measures by targeting specific application's data traffic in order to alleviate the error condition observed on a link. Embodiments of the technology are further directed to monitoring, storage and analysis of information with regards to error causing payload data patterns along with frame data quality parameters to provide a forensic analysis of link connectivity performance across the network. In addition to improving network link utilization and up time, aspects of the technology also provide application level link analytics to extend application continuity (i.e. application movement or application modification). The gained analytics insight may be used to predict and plan corrections of connectivity failures and provide coordinated hardware updates. Various embodiments of the disclosure are discussed in detail

below. While specific implementations are discussed, it should be understood that this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without parting from the spirit and scope of the disclosure.

(11) Network link errors have been generally attributed to physical or logical discontinuities in the transmission of information. Therefore an excessive number of errors, exceeding a certain threshold, have conventionally been indicative of a faulty link and under such conditions a network link is either automatically or manually shut down. Other areas of study are related to software errors and/or misconfigurations that may cause over-utilization of a link, resulting in excessive accumulation of errors and consequent loss or failure of the link. In many previous studies directed at performance of network links for electronic transmission of information the cause of link failure whether identifiable or not has not featured significantly in performance recovery implementation.

(12) Previously proposed remedies are primarily directed to controlling or reducing number of errors in data transmission by utilizing various encoding techniques such as Forward Error Correction (FEC). Such remedial techniques are generally based on the central idea of encoding the message in a redundant way by using an error-correcting code. The redundancy allows the receiver to detect a limited number of errors that may occur anywhere in the message, and often to correct these errors without retransmission. This however is an attempt at reducing network link errors. Moreover, such techniques are applied to the aggregate traffic traversing the link, which may add significant costs to area, power and latency involved in the implementation of it. Furthermore much of the error on a link may go undetected by such techniques. Other strategies may involve duplication of transmission such that two identical data streams are transmitted across interconnecting links. However, in a bi-connected network such as this, each network node or switch requires two transceivers. This technique would double the transmission hardware and may require additional processing cycles at each termination point.

(13) With a move towards multi-tenant data centers and popularity of cloud-based networking environments, aggregate data traffic traversing a network link may be increasingly comprised of several distinct transmission streams associated with multiple source applications. Yet, the error performance or reliability of network links is almost always considered in the context of aggregate traffic traversing the network link. Therefore, whether expressed over time, or instantaneously, or as a rate or count parameter, a network link error/fault condition is almost always determined on the basis of the aggregate errors, and monitored as such. Consequently a link is shut down if observed aggregate errors on the link exceed a pre-determined threshold, thus taking down link connectivity for all source applications that may be transmitting over the link.

(14) One aspect of the present technology puts forth the premise that the behavior of some applications, for example specific bit pattern in a payload of an application data frame, may result in apparent link errors whilst the behavior of other applications may not demonstrate errors. Therefore, according to an embodiment of the present technology, under such conditions relocation of application traffic onto a different transmission path or bit pattern modification of application payload, associated with observed link errors, may result in mitigation of the reported link errors. Therefore, a more granular analysis of link error condition based on differentiating among multiple application-specific transmissions traversing the link may prevent unnecessary link shut down and connectivity failures that may have otherwise been avoided.

(15) As posited above and in accordance to one aspect of the present technology, often times link errors are due to specific data patterns transmitted by a particular source application. Certain bit patterns undergo additional attenuation under non-ideal or noisy transmission link conditions. Such data pattern dependent signal attenuation may lead to bit flips in the payload bit stream thus contributing to the aggregate link error as observed by a terminating device. Therefore certain transmission errors, counted into the aggregate link errors are in fact associated only with a particular transmission sourced from a specific application. Such transmission errors may therefore be unrelated to and not indicative of connectivity performance or error conditions experienced by

other applications that may be transmitting over the common link.

(16) A total link shutdown as a result of excessive errors that may be narrowed to a particular source application is adverse to the overall network performance. In such cases, a more optimal course of action may involve moving the said source application to a different link/path or perhaps notifying the application that is causing that issue to take corrective actions instead of actually shutting down the link which will bring down all the applications transmitting over the link. This becomes particularly important when multiple applications are communicating over a common link.

(17) FIG. 1 illustrates a general data transmission format **102** for communication across a link established, for example, between device **104** and **106**. Transmission format **102** comprises frame delimiters such as Start Of Frame (SOF) indicator **108** and the End Of Frame (EOF) indicator **110** which identify the start and end of a frame, respectively. Transmission format **102** further comprises primitive signals that may be used to indicate events and actions. A primitive signal, corresponding to IDLE frames is denoted by element **112** in the transmission format **102**. IDLE frames may be used to communicate status messages between communicating ports/devices. If no user-data is transmitted the ports may send IDLE frames. These are frames with some bit pattern that may be needed to maintain bit and word synchronization between two adjacent ports/devices. Additionally transmission of IDLE frame may fulfill a required standard corresponding to a minimum number of primitives required between frames.

(18) FIG. 1 further illustrated an exemplary transmission format **114** wherein the IDLE frame **112**, in front of the Start Of Frame indicator **108** is replaced with an application tag **116**. In this way transmission **114** traversing a transmission link between **104** and **106** is tied to a source application and is distinguishable as such. In FIG. 1, exemplary transmission format/frame **114** is overloaded with an inline application Tag **116** in place of the IDLE primitive **112** in front of the SOF indicator **108**. By allowing inline usage of the application tag **116** no extra bandwidth is taken up by the transmission format/frame **114**.

(19) It should be noted that inline insertion of an application tag in place of IDLE frame in a transmission stream, as illustrated in FIG. 1, represents one exemplary method of identifying source application for different transmission streams traversing a link. Embodiments of the present technology may involve other ways of identifying the source application corresponding to each data frame traversing a transmission link without application tag. For example, source application may be inferred from the encapsulated information in the data packet itself by examining any combination of layer 2, layer 3 or layer 4 or user defined payload data. Alternatively, identification of source application may be facilitated with an embedded in packet application tag. Other methods of facilitating differentiation of aggregate traffic over a link on a basis of corresponding source application may be apparent to a person of ordinary skill in the art.

(20) FIG. 2 illustrates an exemplary representation of aggregate traffic profile traversing an application-aware link **200** in accordance to an embodiment of the present technology. The aggregate traffic traversing the application-aware link **200** is differentiated on a basis of source application such that each distinct transmission stream traversing the link is tied to a specific source application as identified by an application tag inserted into each distinct transmission stream. Similarly, on the receiving end, an error monitoring mechanism is independently established for individual transmission streams such that an application specific transmission that may be disproportionally contributing to aggregate link errors may be quickly and accurately identified and appropriate corrective action taken. For example, with reference to FIG. 2, transmission streams from source applications **204**, **206**, **208** and **210** on the transmitting device end **211** of the application-aware link **200**, as respectively identified by application tags **212**, **214**, **216** and **218**, are tracked and monitored by dedicated Bit Error Rate counters **220**, **222**, **224**, and **226** on the receiving device end **228** of the application-aware link **200**.

(21) In some embodiments, application traffic flow may be characterized by a presence of an

application tag or the absence of an application tag to identify performance level of a particular link carrying traffic from one or more source applications.

(22) FIG. 3 illustrates an operational flow diagram **300** directed to an exemplary method of operation for an application-aware link, in accordance to an embodiment of the present embodiments. In accordance to the exemplary method **300**, identification of a specific source application as a primary cause of excessive error condition on or failure of a transmission link may be followed by a set of corrective actions to both maintain link connectivity and initiate collection of forensic and diagnostic information on the said link for immediate or later analysis.

(23) According to the exemplary method **300** in FIG. 3, application-specific traffic is tagged with a source application identifier at step **302**. This may be accomplished, for example, by a transmitting device or host. Step **302** may involve, for example, inline usage of an application tag in the transmission frames corresponding to a specific application. A copy of the transmission packet(s) is then saved by the transmitting device at **304** and the packet is sent over a link to a receiving device/entity at **306**. Subsequent to tagging the traffic, the transmitting device may start to collect frame data quality per each traffic flow associated with a particular application tag, as denoted by step **307** in the operational flow diagram **300**.

(24) At **308** a decision is made, for example by the receiving device, to initiate tracking of the link error performance relative to each distinct application tag in the incoming traffic. The decision at **308** may be based upon observation of a condition such as aggregate link errors exceeding a target threshold. According to one embodiment of the present technology, tracking of link error condition relative to each distinct application tag may be accomplished, for example, by tracking a dedicated bit error counter for each application specific traffic flow as identified by the application tag inserted in each transmission frame. In the exemplary method **300**, link error condition is evaluated on a basis of Bit Error Rate (BER) that may be tracked using a BER counter.

(25) If the outcome of application-specific error monitoring action initiated at **308** determines that condition **309**, corresponding to Bit Error Rate (BER) being less than expected BER of the link, applies to specific application traffic, the operation moves to **310**. At **310** a message comprising delete instruction of the packet(s) stored at **304** is sent to the transmitting device. This message may be inserted into the application tag, for example, corresponding to a format; AppTag=Delete packet stored at step **304**. The stored packet is deleted at step **310** and operation moves back to **302**.

(26) However, if the outcome of application-specific error monitoring action initiated at **308** determines that condition **311**, corresponding to Bit Error Rate (BER) being greater than the fault tolerance of the link, applies to specific application traffic, the operation is moved to step **312**. At **312** receiving entity/device sends a request for retransmission of the packet(s) associated with excessive link to the transmitting device in order to verify repeatability of the error. For example, receiving entity/device may send a message **313** to the transmitting device comprising of |AppTag=Retransmit last packet stored in step **304**|. Upon receipt of the retransmit request, the transmitting device retransmit packet(s) saved in step **304** to the receiving entity/device. At **314** the receiving device will determine the repeatability of the error condition in the retransmitted packet(s). If error condition **316**, corresponding to retransmitted packet(s) having repeatable data errors is confirmed, the receiving device will send a message (i.e., AppTag=Repeatable Error) to the transmitting device to notify it about the observed repeatable excessive error condition. The receiving device deletes the received retransmitted packet and does not forward it. At this point the operation moves to **318** wherein the transmitting device moves the complete frame stored at step **304** and frame quality data from step **307**, as indicated by **319**, to a scratch space quarantined for analysis by a collector and marked as repeatable. The Collector may alert the application of repeatable error with high BER and requests the application to take a different path (i.e., alternative server Network Interface Card (NIC) if dual homed or move to a different Virtual Machine/Container.

(27) If error condition **320**, corresponding to retransmitted packet not having repeatable data errors

is confirmed, the operation moves to **322** wherein the receiving device/entity will send a message to the transmitting device to notify it about the non-repeating error condition. Such a message, for example, may comprise AppTag=Not Repeatable Error. At this point the operation may move back to step **310**, corresponding to the deletion of frame(s) stored by the transmitting device at step **304**, or it may move to **318** wherein complete frame(s) from step **304** and frame quality data from step **307** stored for each application tag is moved to a scratch space quarantined for analysis by collector and marked as not repeatable.

(28) In some embodiments, if a collector alerts a transmitting device (or a source application directly) of unrepeatable high BER condition, source application may choose or be directed to take a different path or it may treat the message as an informational event, depending on resource availability. Receiving entity/device may then delete the retransmitted packet and return to step **302**, as indicated by **324**. The decision to relocate application-specific traffic to a different path or modify application payload in response to repeatable high error condition message or to treat a non-repeatable high error condition message as informational may be made by the transmitting device, source application or a different control program.

(29) It should be noted that a bit error rate is just one measurement of error condition. Many different ways of quantifying error performance, such as CRC error, checksum errors, etc., may be deployed as a performance measure for a transmission link without departing from the scope and spirit of the present technology.

(30) It should also be noted that the window of error measurement may be indefinitely long living, for example, it may be cleared after a programmable duration where the application tag is no longer seen (i.e., stale entry), or it may be cleared after a fix user programmable duration regardless of activity. The criteria may be determined in accordance to the requirements of an application or the intended purpose of a user/administrator.

(31) In some embodiments quarantined data may be logged in NVRAM or exported to collector for Quality Assurance reproduction, debug and fix.

(32) In some embodiments, the retransmit request, for error repeatability verification, sent by a receiving entity/device may involve retransmission of stream of N packets. In some embodiments, examination of application-specific traffic streams for the determining the extent of error contribution from individual source application may be initiated upon link error condition exceeding an error tolerance of a particular transmission link. Alternatively, application-specific traffic error monitoring may be initiated/triggered in accordance to other relevant performance parameters, or in response to an occurrence of an external trigger, derivation of histogram, average and other statistical charts, or a user specified condition. It is understood that application-specific traffic error monitoring may be initiated/triggered in accordance to a variety of criteria as would be apparent to a person of ordinary skill in the arts.

(33) In some embodiments of the present technology, frame data quality may comprise parameters such as temperature, payload pattern, voltage, time of occurrence, cable manufacture, aging, wear and tear, stress, humidity, location, altitude, etc. Preserving of exact data payload and frame quality parameters is valuable for reproducing, debugging and fixing link faults and connectivity issues. In some embodiments, heuristics/machine learning techniques may be applied to collected information on faulty application payload data patterns implicated in link failures in conjunction along with collected frame quality data parameters (that may include location/height/humidity/temperature information) to thereby provide an effective analytics-based tool for discovery of physical resource problems impacting application traffic, while also achieving high link utilization.

(34) As previously described, quality of a transmission link may be impacted by such factor as, age and wear of hardware, environmental conditions to which link hardware or traversing signals may be subject (i.e., temperature and/or voltage) and in certain cases the receivers on either side or along the transmission link. Certain bit patterns in the payload of an application data frame may be

more susceptible to experiencing signal attenuations, during the transition across a link subject to the aforementioned conditions. This bit-pattern dependent attenuation may result in one or more bit flips in the payload of application data frames thus causing link errors that to varying degrees may bring down the average or aggregate error rate of the link. For example, a connectivity protocol such as Ethernet as has an allowed error tolerance that corresponds to $10e-12$ bit errors in a second (BER of $10.\text{sup.}-12$). Therefore, according to an embodiment of the present technology, when observed rate of bit errors on an Ethernet link exceeds the aforementioned threshold amount, inspection of application specific traffic may be triggered in order to determine if the observed error condition is caused by particular application data traffic. By utilizing metrics that are based on determination of average error over time such as BER or standard deviation of error, an application's throughput rate or size of the payload in a transmission frame will not factor into the quantification of the error condition on the link.

(35) FIG. 4 represents an exemplary Bit Error Rate map **400** used for illustrating a performance of transmission links in exemplary network **402**. In the exemplary Bit Error Rate map **400**, performance of different transmission links are indicated by a Bit Error Rate (BER) parameter. Furthermore, for illustration purposes, link quality, ranging from Exceptional to Poor, is indicated by the width of the line representing a link. If the exemplary network **402** is taken to represent an Ethernet network, and the link error threshold condition is set according to Ethernet link fault tolerance, which in terms of BER may be represented as $1 \times 10.\text{sup.}-12$ then Ethernet links **404**, **406**, **408** and **410** with BER levels far below Ethernet BER threshold of $1 \times 10.\text{sup.}-12$, are indicated as exceptional. Ethernet links **412** and **414** with BER levels right at the Ethernet BER tolerance threshold are indicated as acceptable. Ethernet link **416** with BER moderately below Ethernet BER tolerance threshold is indicated as marginal and Ethernet links **418** and **420** with BER far above fault tolerance threshold of Ethernet are indicated as poor.

(36) In reference to exemplary network **402**, under BER conditions as annotated in FIG. 4, existing standards would call for shut down and subsequent replacement of links **418**, **420** and **416** due to excessive BER. However, applying techniques and methods disclosed in accordance with some embodiments of the present technology, to network **402** may significantly improve the outcome in terms of network utilization and up time. This may be better illustrated through expanded view **425** representing link **418** as it would be implemented under application-aware error management and recovery techniques disclosed in accordance to embodiment of the present technology.

(37) In accordance to embodiments of the present technology, a BER of $1 \times 10.\text{sup.}-05$ on Ethernet link **418** established between devices **422** and **424** may never be observed because the condition will be corrected as the BER exceeds the tolerance threshold of the link, which in case of Ethernet link **418** would correspond to $1 \times 10.\text{sup.}-12$. As illustrated in **425**, under disclosed embodiments of the present technology, aggregate traffic traversing link **418** may be differentiated as transmission streams **426**, **428** and **430** associated with source applications **432**, **434** and **436**, respectively. As the observed BER on link **418** exceeds the link's failure threshold **437** (for example as it exceeds BER level of $1 \times 10.\text{sup.}-12$ or perhaps as it reaches a BER level of $1 \times 10.\text{sup.}-11$, in case of an Ethernet link) application level BER monitoring of data traffic across **418** will be initiated to thereby track link errors associated with each individual application traffic as identified by a corresponding application tag.

(38) With reference to view **425** of transmission link **418**, application level tracking of link errors will identify traffic flow **426** sourced from application **432** as the primary cause of failure for link **418**. As such corrective actions to alleviate the error condition on the link will be directed at the application **432** instead of involving the entire link which would disrupt traffic from non-offending applications. The corrective action may comprise moving application **432** to a different virtual or physical device or rerouting application traffic **426** to take a different path to **424**. Corrective action may also include modifying the bit pattern associated with application traffic **426** in such a way so as to remove the resulting error condition. In either case without contribution of the offending

application traffic **426** the condition of the link will improve immediately to exceptional or at worst acceptable as indicated by the link quality index in FIG. **4**.

(39) If application-aware error monitoring and recovery, in accordance to some embodiments of the present technology, is not implemented in case of the exemplary network **402**, error conditions commensurate with annotated BER levels will result in swapping or replacement of link **418**, which would unnecessarily take down connectivity for applications **436** and **434**. This situation would also apply to links **420** and **416** having BER in excess of the allowed threshold. However, if an application-aware error monitoring and recovery algorithm, in accordance to embodiments of the present technology, is implemented, transmission links burdened by specific application traffic, such as **418**, would recover automatically.

(40) In some embodiments, a plurality of application tags may be used in combination, or one or more application tags may be used in combination with user defined payload data to enable a more granular combination and filtering of the traffic (i.e., application tag maybe used in combination with hypertext transfer protocol port numbers.)

(41) In some embodiments, the application tag may be inline and grow the data frame rather than preceding the Start of Frame indicator if the user so desires as shown in FIG. **5**. In referencing FIG. **5**, a packet/frame may have more than one application tag that may replace IDLE primitive(s) as shown by exemplary instance **502**, or inserted within the frame as shown by exemplary instance **504**.

(42) Returning back to FIG. **5**, in case of inline referring to a single packet, the associated tag may be a carryover (derived) by a previous link or a collection of N application tags derived from M previous links. For example, in the exemplary instance **506** illustrated in FIG. **5**, application tag **508** in packet/frame **510** traversing link **512** is carried over (derived) from a previous link **513**. Similarly collection of three application tags **514**, **516**, and **518** in packet/frame **520** traversing link **522** constitute carryovers from two previous links **512** and **513**.

(43) Selection of one or more N application tags may be used to update profile measurement of one or more error rates profiles as shown by exemplary depiction **602** in FIG. **6**. **602** comprise a block **604** on the receiving side of the link **606**. Block **604** reflects that multiple application specific traffic flows (as expressed by multiple application identifiers/tags) may be tracked (instantly or historically) for quality performance (i.e., error performance) depending on the number application identifiers present. As further illustrated by the exemplary representation **610** in FIG. **6**, the plurality of applications identifiers may be nested, overlapping cascaded, added and/or replaced. For example, in referencing example **610** in FIG. **6**, application tags **612** and **614** are added over the links **616** and **618** respectively. Application tag **614** is then deleted over link **620** while application tags **622** and **624** are added.

(44) In some embodiments, Application tags corresponding to low error rate traffic flows (i.e., excellent Bit Error Rate) may be monitored and identified for the purpose of providing feedback to relevant operators that a link can be used for high priority, high importance, high value traffic due to the low error rate.

(45) Disclosed system and method, in accordance to some embodiments of the present technology, in addition to increasing link utilization and up time across the entire network, will also facilitate a historical analysis of the link transitions across various quality grading (i.e., exceptional, poor, acceptable, marginal). This information, combined with an analysis of corresponding frame data and frame data quality parameters may facilitate a forensic understanding of network wide physical resource issues and link performance profile affecting application traffic.

(46) In some embodiments, in scenarios corresponding to multiple traffic streams failing over a transmission link, a self-test mechanism may be deployed on the link to separately verify a link's error performance or quality prior to swapping a link or relocating all the link traffic. In some embodiments, self-test mechanism may involve transmission of a diagnostic test pattern over the link to verify existence of application independent error conditions on the link.

(47) Some embodiments of the present technology are directed at a system and method for mitigating application payload related link failure. This may be accomplished by monitoring errors for each application specific transmission flow traversing a link. Identifying different application specific transmission flows maybe accomplished by several different means namely, by making inferences based on an inspection of the information encapsulated in a packet. One exemplary method disclosed by the present technology includes insertion of an application tag inline with the data frame (transmission frame). In one embodiment, an inline application tag may replace an IDLE primitive in front of the SOF field of the frame. In this way no extra bandwidth is used up as a result of application tagging of data frames. Alternatively, application tagging of data frames may be accomplished by using an embedded in packet application tag instead of an inline application tag. Aspects of the present technology are directed to identification of distinct traffic streams traversing a link to thereby evaluate its relation to one or more error conditions observed on the link

(48) According to some aspects of the technology, once different transmission streams traversing the link are individually identified and monitored based on the corresponding source application, contributions of different application specific transmissions to the overall error condition observed on the link may be determined and corrective actions may be taken in response to the one or more application traffic identified as a main source of observed excessive errors resulting in link failure.

(49) Excessive error contribution from a particular application traffic stream may be due to specific payload data pattern propagating over a link with non-ideal attributes or conditions. Once an application traffic stream that is disproportionately contributing to an observed link error condition is identified, corrective action may be taken to alleviate the error condition without having to take down the link. In one embodiment, corrective action may include an initial request for retransmission of the offending packet or stream of N packets to determine repeatability of the error condition. The initial request for retransmission may be sent by a receiving entity to a transmitting entity, in accordance to one embodiment. Other embodiments may involve an initial request for retransmission sent directly to a source application or to a separate management and control program.

(50) When one or more specific transmissions are identified and confirmed as a cause of a link failure, subsequent corrective actions, as disclosed by one embodiment of the present technology, may include relocating application to a different server/Virtual Machine/container or updating relevant routing protocols to re-route/relocate the application traffic onto a different path, thereby keeping the link available for other transmission streams traversing the link.

(51) In some embodiments, corrective actions may include modifying the payload data pattern of the one or more application specific data transmissions or recommending application data modification in such a way so as to improve a quality of propagation signal associated with the specific transmission. In other embodiments, corrective action may further include monitoring, storage and analysis of information associated with offending payload data patterns along with frame quality data parameters such as temperature, voltage, humidity, altitude, hardware manufacturers, etc. This information may be used to provide link analytics to facilitate prediction and planning of remedial or corrective actions.

(52) According to some embodiments, a collector may be utilized to monitor, collect and process information associated with payload data patterns and frame data quality parameters to provide analytics insight. A collector, in accordance to one aspect of the embodiments, may also be used to influence application payload based on acquired knowledge of learned error data pattern. For example, a continuous stream of constant string of “zero” value reserved or padding payload fields can increase noise of a physical link such as a cable. That application if alerted can alternatively replace the cable noisy (i.e. reserved or padding) data with harmless random values; hence, immunize application from cable weaknesses.

(53) According to some embodiments of the technology, collection of frame data quality and other

forensic parameters may be triggered by application aware link errors and processed to provide exact location along with performance and condition characteristics of cloud-based or in-house data center. This information may then be dynamically disseminated to relevant operators to alert them of the exact location of cloud datacenter performance and condition characteristics without polling or continuous parameter streaming. According to one embodiment, edge computing may be used to ensure collector database receives useful information.

(54) A link failure may be characterized as a condition whereby an error condition exceeds a predetermined threshold. One embodiment implements Bit Error Rate (BER) as a parameter to indicate and quantify observed error condition on a transmission link. However, in alternative embodiment, BER may be replaced by other performance metrics such as minimum, maximum, average or standard deviation error counts. In some embodiments, error conditions may correspond to one or more user defined conditions or criteria.

(55) For clarity of explanation, in some instances the present technology may be presented as including individual functional blocks including functional blocks comprising devices, device components, steps or routines in a method embodied in software, or combinations of hardware and software.

(56) In some embodiments the computer-readable storage devices, mediums, and memories can include a cable or wireless signal containing a bit stream and the like. However, when mentioned, non-transitory computer-readable storage media expressly exclude media such as energy, carrier signals, electromagnetic waves, and signals per se.

(57) Methods according to the above-described examples can be implemented using computer-executable instructions that are stored or otherwise available from computer readable media. Such instructions can comprise, for example, instructions and data which cause or otherwise configure a general purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. Portions of computer resources used can be accessible over a network. The computer executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, firmware, or source code. Examples of computer-readable media that may be used to store instructions, information used, and/or information created during methods according to described examples include magnetic or optical disks, flash memory, USB devices provided with non-volatile memory, networked storage devices, and so on.

(58) Devices implementing methods according to these disclosures can comprise hardware, firmware and/or software, and can take any of a variety of form factors. Examples of such form factors include laptops, smart phones, small form factor personal computers, personal digital assistants, and so on. Functionality described herein also can be embodied in peripherals or add-in cards. Such functionality can also be implemented on a circuit board among different chips or different processes executing in a single device, by way of further example.

(59) The instructions, media for conveying such instructions, computing resources for executing them, and other structures for supporting such computing resources are means for providing the functions described in these disclosures.

(60) Although a variety of examples and other information was used to explain aspects within the scope of the appended claims, no limitation of the claims should be implied based on particular features or arrangements in such examples, as one of ordinary skill would be able to use these examples to derive a wide variety of implementations. Further and although some subject matter may have been described in language specific to examples of structural features and/or method steps, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to these described features or acts. For example, such functionality can be distributed differently or performed in components other than those identified herein. Rather, the described features and steps are disclosed as examples of components of systems and methods within the scope of the appended claims.

Claims

1. A method comprising: tagging, by a transmitting device, application-specific traffic from an application with a source application identifier which identifies the application; transmitting, by the transmitting device, network traffic, including the tagged application-specific traffic to a receiving device; saving, by the transmitting device, a copy of the application-specific traffic that was tagged and transmitted; initiating, by a receiving device, monitoring of link error conditions relative to the tagged application-specific network traffic; and in response to the monitoring the link error conditions, creating and sending, by the receiving device, a message with a tag associated with the monitoring to the transmitting device that tagged the application-specific traffic.
2. The method of claim 1, wherein the tagging includes adding one or more application tags inside one or more data frames corresponding to a specific application.
3. The method of claim 1, wherein the monitoring includes determining whether a link error condition exceed a target threshold.
4. The method of claim 3, wherein the link error condition at least includes a bit error rate tracked using a counter.
5. The method of claim 1, wherein in response to a link error condition of the link error conditions indicating a threshold is not met, creating and sending the message with a delete tag, wherein the delete tag indicates deleting of a least one packet of the copy of the application-specific network traffic.
6. The method of claim 1, wherein in response to a link error condition of the link error conditions indicating a threshold is met, creating and sending the message with a retransmission tag, wherein the retransmission tag indicates retransmitting of a least one packet of the copy of the application-specific network traffic.
7. The method of claim 6, further comprising: in response to retransmission of the least one packet of the application-specific network traffic, determining a repeatability of the link error condition; and creating and sending a message with a repeatable error tag, wherein the repeatable error tag indicates observed repeatable excessive error condition.
8. One or more non-transitory computer-readable media storing computer-executable instructions that, when executed by one or more processors, cause the one or more processors to: tag, by a first processor, application-specific traffic from an application with a source application identifier which identifies the application; transmit, by the first processor, network traffic, including the tagged application-specific traffic to a receiving device; save, by the first processor, a copy of the application-specific traffic that was tagged and transmitted; initiate, by a second processor, monitoring of link error conditions relative to the tagged application-specific network traffic; and in response to the monitoring the link error conditions, create and send, by the second processor, a message with a tag associated with the monitoring to the the first processor that tagged the application-specific traffic.
9. The one or more non-transitory computer-readable media of claim 8, wherein the tagging includes adding one or more application tags inside one or more data frames corresponding to a specific application.
10. The one or more non-transitory computer-readable media of claim 8, wherein the monitoring includes determining whether a link error condition exceeds a target threshold.
11. The one or more non-transitory computer-readable media of claim 10, wherein the link error condition at least includes a bit error rate tracked using a counter.
12. The one or more non-transitory computer-readable media of claim 8, wherein in response to a link error condition of the link error conditions indicating a threshold is not met, the tag is a delete tag that indicates deleting of a least one packet of the copy of the application-specific network traffic.

13. The one or more non-transitory computer-readable media of claim 8, wherein in response to a link error condition of the link error conditions indicating a threshold is met, the tag is a retransmission tag that indicates retransmitting of a least one packet of the copy of the application-specific network traffic.
 14. The one or more non-transitory computer-readable media of claim 13, comprising further instructions that, when executed by the one or more processors, cause the one or more processors to: in response to retransmission of the least one packet of the application-specific network traffic, determining a repeatability of the link error condition; and creating and sending a message with a repeatable error tag, wherein the repeatable error tag indicates observed repeatable excessive error condition \.
 15. A system comprising: a transmitting device configured to: tag application-specific traffic from an application with a source application identifier which identifies the application; transmit network traffic, including the tagged application-specific traffic to a receiving device; and save a copy of the application-specific traffic that was tagged and transmitted; a receiving device configured to: initiate monitoring of link error conditions relative to the tagged application-specific network traffic; and in response to the monitoring the link error conditions, create and send a message with a tag associated with the monitoring to the transmitting device that tagged the application-specific traffic.
 16. The system of claim 15, wherein the tagging includes adding one or more application tags inside one or more data frames corresponding to a specific application.
 17. The system of claim 15, wherein the monitoring includes determining whether a link error condition exceeds a target threshold.
 18. The system of claim 15, wherein in response to a link error condition of the link error conditions indicating a threshold is not met, the tag is a delete tag that indicates deleting of a least one packet of the copy of the application-specific network traffic.
 19. The system of claim 15, wherein in response to a link error condition of the link error conditions indicating a threshold is met, the tag is a retransmission tag that indicates retransmitting of a least one packet of the copy of the application-specific network traffic.
 20. The system of claim 18, wherein the receiving device is further configured to: in response to retransmission of the least one packet of the application-specific network traffic, determining a repeatability of the link error condition; and creating and sending a message with a repeatable error tag, wherein the repeatable error tag indicates observed repeatable excessive error condition.
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